High-Resolution Direct Position Determination Using MVDR

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***Abstract*— The direct position determination (DPD) approach is a single-step method, which uses the maximum likelihood estimator to localize sources emitting electromagnetic energy using combined data from all available sensors. The DPD is known to outperform the traditional two-step methods under low signal-to-noise ratio conditions. We propose an improvement to the DPD approach, using the well-known minimum-variance-distortionless-response (MVDR) approach. Unlike maximum likelihood, the number of sources needs not be known before applying the method. The combination of both the direct approach and MVDR yields unprecedented localization accuracy and resolution for weak sources. We demonstrate this approach on the problem of multistatic radar, but the method can easily be extended to general localization problems.**

***Index Terms*— Positioning algorithms, adaptive array processing, minimum variance distortionless response (MVDR), maximum likelihood estimation, direct position determination.**

NOMENCLATURE

$(.)^T$	Transpose.
$(.)^H$	Conjugate transpose.
$\|\mathbf{a}\|$	Euclidean norm of a vector.
diag $(\mathbf{A})$	Main diagonal of a matrix.
$\otimes$	Kronecker product.

I. INTRODUCTION

THE localization of sources emitting electromagnetic or acoustic energy is needed in wild-life tracking, radio-astronomy, seismology, medical-diagnosis, communications, and other engineering applications. Common localization methods use two estimation steps. First, intermediate parameters are estimated. Intermediate parameters are usually time of arrival, direction of arrival, Doppler frequency shift or signal strength. These estimated parameters are then used, in a second step, to estimate the actual location of the emitter. The Direct Position Determination (DPD) approach has been recently proposed [1], [2] as a single-step Maximum Likelihood localization technique. A single-step approach is a technique in which the estimator uses exactly the same data as used in two-step methods but estimates the source location directly, skipping the intermediate (first) step. This can be viewed as searching for the emitter location that best explains the collected data. From estimation theory point-of-view the two-step approach is inferior, since in the first step the parameters are measured independently, ignoring the constraint that the measurements relate to the same emitter location. Indeed, the DPD method has been shown to be superior to the two-step methods for low SNR. In addition, this method has been shown to be more robust by inherently selecting reliable observations without the need for a goodness-of-fit test (such as the chi-square test). This method was also extended to radar scenarios in [3], where the Maximum Likelihood target location estimation was developed as well as the Cramer Rao lower bound for the estimation error.

When there are multiple sources the DPD is no longer equivalent to the Maximum Likelihood Estimator (MLE). The exact MLE can be derived but it requires a multi-dimensional search which is usually impractical. An alternative for the Maximum Likelihood parameter estimator is the Minimum Variance Distortionless Response (MVDR) estimator. It was originally proposed by Capon [4] for frequency-wavenumber power spectral density analysis, but has since been used extensively as a high resolution method. The idea is to maximize a score function given by multiplying the inputs by a set of weight vectors, where the weight vectors are adaptively selected in order to fix the response for the parameter value of interest while minimizing the output power. Unlike the Maximum Likelihood approach, the MVDR approach does not need to know a-priori the number of targets (or model order) and therefore it is a robust approach with good resolution and immunity to jamming and interference. The MVDR was proposed previously for Synthetic Aperture Radar (SAR) high resolution imaging. DeGraff [5] describes many approaches for improving SAR images including MVDR which he terms MVM for Minimum Variance Method. According to [5] MVM is one of the most successful approaches.

Our test case is multistatic radar, thus a brief introduction is presented here. The idea of bistatic radar has been studied from the earliest days of radars. By spatially separating the transmitter from the receiver, these radar systems have been shown to exhibit some advantages over common monostatic radar systems. A multistatic radar system is an extension of the bistatic radar concept, where multiple cooperative receivers are used for target localization. This could be generalized further with the addition of multiple transmitters (a scheme usually termed MIMO radar, e.g. [7]–[9]), but we limit our

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demonstration to a single transmitter, thus ignoring difficulties caused by mutual interference of the transmitted signals.

The focus of this paper is the demonstration of the single-step (direct) MVDR concept for source localization. As an example, we use a simplified multi-static radar model, neglecting radar clutter for brevity. Future work includes the full blown radar model. Note that MVDR is applied to the target's location estimation in a single step, and not for intermediate parameters estimation. Some previous publications touch upon our proposed approach but in no way cover the full potential of this method (See for example [10], where only the signals' direction of arrivals are taken into account, and [11]–[13] where the receivers are collocated). Some papers like [14] mention "location" in their titles but their text discuss application of MVDR to Angle of Arrival. We show that the estimation method proposed here can significantly improve target resolution compared with the single-source MLE. Fine target resolution can prove very useful for target localization within many dense decoys.

Our novel contributions are summarized as follows. 1) We provide a 1-step MVDR based localization algorithm and demonstrate its unprecedented resolution capabilities. MVDR is applied to localization and not, as done in previous works, to intermediate parameters, where actual localization is performed in a second step. 2) We develop a way to predict estimation performance for a single target at high SNR based on small-error analysis. 3) The localization problem is presented in a way which easily lends itself to the application of other single-step Beamformers.

In the sequel the terms DPD and Maximum Likelihood refer to the Maximum Likelihood Estimator assuming a single source, also known as conventional Beamforming, and not to the exact Maximum Likelihood, which needs to know (or guess) a-priori the number of sources, and which requires a multi-dimensional search that is often impractical.

II. Problem Formulation

Consider a transmitter and L widely separated receiving arrays. Each receiving array consists of M_r elements. The array aperture is typically a few signal wavelengths. The transmitted signal consists of several consecutive pulses, where the waveform of each pulse, $s(t)$, is confined to the time interval $t \in [0, T]$. It is assumed that the signal $s(t)$ is perfectly known, which is usually the case when there is line of sight from the transmitter to the receiving array. In the following analysis the signal impinges on a single target whose coordinates vector is denoted by $\mathbf{p}_t$, and is reflected by the target towards the receiving arrays. We assume that the transmitter, arrays and target are all confined to a plane, and the transmitter and receiving arrays locations are known while the target location needs to be estimated.

The ℓ-th array output for a single pulse is given by the $M_r \times 1$ vector,

$$\mathbf{r}_\ell(t) = \mathbf{a}_\ell(\mathbf{p}_t)\alpha_\ell s(t - \tau_\ell(\mathbf{p}_t))e^{i2\pi f_{D,\ell} t} + \mathbf{n}_\ell(t) \tag{1}$$

where α_ℓ is the signal attenuation at the ℓ-th array, τ_ℓ is the signal delay associated with the propagation from the transmitter to the target and then to the ℓ-th array, which satisfies,

$$\tau_\ell(\mathbf{p}_t) = \frac{\|\mathbf{p}_t - \mathbf{p}_\ell\|}{c} + \frac{\|\mathbf{p}_t - \mathbf{p}_{\mathrm{Tx}}\|}{c} \tag{2}$$

where $\mathbf{p}_{\mathrm{Tx}}$ is the transmitter's location, $\mathbf{p}_\ell$ is the ℓ-th array location, and c is the speed of propagation. Further, $\mathbf{n}_\ell(t)$ is a $M_r \times 1$ wide-sense stationary, white, zero mean, complex Gaussian noise, $\mathbf{a}_\ell(\mathbf{p}_t)$ is a $M_r \times 1$ vector representing the ℓ-th array response to a target at $\mathbf{p}_t$, and $f_{D,\ell}$ is the Doppler frequency shift. We note that without loss of generality we can impose the constraint $\|\mathbf{a}_\ell(\mathbf{p}_t)\|^2 = 1$. Since the transmitter and arrays cooperate the signal transmission time is perfectly known. This can be accomplished by direct interception of the transmitted signal or by synchronization of the transmitter and receiving arrays. Finally, we assume the target is illuminated by M_p consecutive pulses. To simplify the exhibition, it is assumed that the target speed is small enough to neglect the Doppler effect. This assumption would be addressed in Sec. III. The DFT of the received j-th pulse is given by

$$\bar{\mathbf{r}}_{\ell,k}(j) = \mathbf{a}_\ell(\mathbf{p}_t)\,\alpha_\ell(j)\,\bar{s}_k e^{-i2\pi f_k \tau_\ell(\mathbf{p}_t)} + \bar{\mathbf{n}}_{\ell,k}(j) \tag{3}$$

where $f_k = \frac{k}{K} f_s$ is the frequency associated with the k-th coefficient, K is the number of samples, f_s is the sampling frequency, and $\bar{\mathbf{r}}_{\ell,k}$, $\bar{s}_k$ and $\bar{\mathbf{n}}_{\ell,k}$ are the k-th Fourier coefficients of $\mathbf{r}_\ell(t)$, $s(t)$ and $\mathbf{n}_\ell(t)$, respectively, and where it is assumed that the observation time is longer than the received signal interval plus its delays at all sensors.

Define

$$\begin{aligned}
\bar{\mathbf{r}}_\ell(j) &\triangleq [\bar{\mathbf{r}}_{\ell,1}^T(j), \bar{\mathbf{r}}_{\ell,2}^T(j), \ldots, \bar{\mathbf{r}}_{\ell,K}^T(j)]^T \\
\bar{\mathbf{n}}_\ell(j) &\triangleq [\bar{\mathbf{n}}_{\ell,1}^T(j), \bar{\mathbf{n}}_{\ell,2}^T(j), \ldots, \bar{\mathbf{n}}_{\ell,K}^T(j)]^T \\
\mathbf{A}_\ell(\mathbf{p}_t) &\triangleq \mathrm{diag}(e^{-j2\pi f_1 \tau_\ell}, \ldots, e^{-j2\pi f_K \tau_\ell}) \otimes \mathbf{a}_\ell(\mathbf{p}_t) \\
\bar{\mathbf{s}} &\triangleq [\bar{s}_1, \ldots, \bar{s}_K]^T
\end{aligned} \tag{4}$$

where the dependence of τ_ℓ on $\mathbf{p}_t$ is suppressed, $\bar{\mathbf{r}}_\ell$ and $\bar{\mathbf{n}}_\ell$ are vectors of size $M_r K \times 1$, $\mathbf{A}_\ell$ is a matrix of dimensions $M_r K \times K$ and $\bar{\mathbf{s}}$ is a vector of size $K \times 1$. We can now write (3) in a vector form

$$\bar{\mathbf{r}}_\ell(j) = \alpha_\ell(j)\,\mathbf{A}_\ell(\mathbf{p}_t)\,\bar{\mathbf{s}} + \bar{\mathbf{n}}_\ell(j). \tag{5}$$

In the next section we derive the (single-source) Maximum Likelihood estimator for $\mathbf{p}_t$, where $\{\alpha_\ell(j)\}$ are treated as unknown parameters. As explained in the introduction, it is possible to derive an exact Maximum Likelihood estimator. However, such an estimator needs to know a-priori the number of targets, and it requires a multi-dimensional search. In Sec. II-B we use the single-source Maximum Likelihood estimator to obtain the multi-source MVDR estimator, which is our main goal.

A. Target Localization Using the Maximum Likelihood Estimator

Using (5) and the fact that $\{\bar{\mathbf{n}}_\ell(j)\}$ are statistically independent complex Gaussian vectors, the Maximum Likelihood cost

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function is

$$Q(\mathbf{p}_t) = \sum_{\ell=1}^{L}\sum_{j=1}^{M_p} \|\bar{\mathbf{r}}_\ell(j) - \alpha_\ell(j)\,\mathbf{A}_\ell(\mathbf{p}_t)\,\bar{\mathbf{s}}\|^2. \tag{6}$$

The signal attenuation $\{\alpha_\ell(j)\}$ is assumed to be independent from pulse to pulse, as is suggested by the well known Swerling II and IV target models, which assume the target radar-cross-section (RCS) is independent from pulse to pulse (see [15]). This assumption will be revisited in Sec. III. The attenuation coefficient that minimizes the cost function (6) is given by

$$\begin{aligned}\hat{\alpha}_\ell(j) &= (\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\,\mathbf{A}_\ell(\mathbf{p}_t)\,\bar{\mathbf{s}})^{-1}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\,\bar{\mathbf{r}}_\ell(j)\\ &= (\bar{\mathbf{s}}^H\bar{\mathbf{s}})^{-1}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\,\bar{\mathbf{r}}_\ell(j)\end{aligned} \tag{7}$$

where we used

$$\mathbf{A}_\ell^H(\mathbf{p}_t)\,\mathbf{A}_\ell(\mathbf{p}_t) = \mathbf{I}_{M_rK}||\mathbf{a}_\ell||^2 = \mathbf{I}_{M_rK}. \tag{8}$$

Substituting (7) back into (6) we get

$$\begin{aligned}&Q_{\mathrm{ML}}(\mathbf{p}_t)\\ &= \sum_{\ell=1}^{L}\sum_{j=1}^{M_p}\left\|\,\bar{\mathbf{r}}_\ell(j) - \epsilon^{-1}\mathbf{A}_\ell(\mathbf{p}_t)\,\bar{\mathbf{s}}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\,\bar{\mathbf{r}}_\ell(j)\,\right\|^2\\ &= \sum_{\ell=1}^{L}\sum_{j=1}^{M_p}\left(\|\bar{\mathbf{r}}_\ell(j)\|^2 - \epsilon^{-1}\left|\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\,\bar{\mathbf{r}}_\ell(j)\right|^2\right)\end{aligned} \tag{9}$$

where

$$\epsilon \triangleq \bar{\mathbf{s}}^H\bar{\mathbf{s}}. \tag{10}$$

Since $\|\bar{\mathbf{r}}_\ell(j)\|^2$ is constant, minimization of this expression is equivalent to the maximization of

$$\begin{aligned}\widetilde{Q}_{\mathrm{ML}}(\mathbf{p}_t) &= \frac{1}{M_p}\sum_{\ell=1}^{L}\sum_{j=1}^{M_p}\epsilon^{-1}\left|\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\,\bar{\mathbf{r}}_\ell(j)\right|^2\\ &= \sum_{\ell=1}^{L}\epsilon^{-1}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\,\hat{\mathbf{R}}_\ell\mathbf{A}_\ell(\mathbf{p}_t)\,\bar{\mathbf{s}}\end{aligned} \tag{11}$$

where the sample covariance is a $M_rK \times M_rK$ matrix, defined as

$$\hat{\mathbf{R}}_\ell \triangleq \frac{1}{M_p}\sum_{j=1}^{M_p}\bar{\mathbf{r}}_\ell(j)\bar{\mathbf{r}}_\ell^H(j). \tag{12}$$

Note that it is always possible, through proper normalization of the received signals, to set $\epsilon = \bar{\mathbf{s}}^H\bar{\mathbf{s}} = 1$.

Finally, we can express the ML estimator as

$$\widehat{\mathbf{p}}_t^{\mathrm{ML}} = \arg\max_{\mathbf{p}_t}\widetilde{Q}_{\mathrm{ML}}(\mathbf{p}_t) = \arg\max_{\mathbf{p}_t}\sum_{\ell=1}^{L}Q_\ell^{\mathrm{ML}}(\mathbf{p}_t) \tag{13}$$

where the individual cost functions are given by

$$Q_\ell^{\mathrm{ML}}(\mathbf{p}_t) \triangleq \left(\mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t)\right)^H\hat{\mathbf{R}}_\ell\mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t) \tag{14}$$

and

$$\mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t) \triangleq \mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}} \tag{15}$$

is a $M_rK \times 1$ weight vector. This ends the derivation for the Maximum Likelihood estimator of the target position assuming a single target and no other interference. This expression can be seen as focusing on the hypothesized target location using the weight vectors $\left\{\mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t)\right\}$, then combining the responses.

Algorithm 1 Target Localization Using Maximum Likelihood

Evaluate $\{\hat{\mathbf{R}}_\ell\}$ according to (12)
Define the area of interest and determine a suitable grid of locations $\mathbf{p}_1,\mathbf{p}_2\ldots\mathbf{p}_G$, where G is the number of grid points
for $j = 1$ to G **do**
  Set $Q(\mathbf{p}_j) = 0$
  for $\ell = 1$ to L **do**
    Evaluate $\tau_\ell(\mathbf{p}_j)$ according to (2)
    Evaluate $\mathbf{A}_\ell\left(\mathbf{p}_j\right)$ according to (4)
    Evaluate $\mathbf{w}_\ell^{\mathrm{ML}}\left(\mathbf{p}_j\right)$ according to (15)
    Let $Q\left(\mathbf{p}_j\right) = Q\left(\mathbf{p}_j\right) + \left(\mathbf{w}_\ell^{\mathrm{ML}}\right)^H\hat{\mathbf{R}}_\ell\mathbf{w}_\ell^{\mathrm{ML}}$
  end for
end for
Find the grid point for which Q is the biggest. This grid point is the estimated position.

Note that the estimator involves the maximization of a nonlinear cost function. A possible algorithm, which uses a simple grid search, is displayed in Algorithm 1. However, it is possible to use more sophisticated algorithms.

B. Target Localization Using MVDR

We now turn to the derivation of the MVDR estimator for this case. As discussed in the introduction, high resolution DOA (Direction of Arrival) or frequency-wavenumber estimation can be achieved by replacing the Maximum Likelihood weight vector with a weight vector designed to minimize the array output power while maintaining a constant response for the hypothesized parameter value. The same idea is implemented here for the target's position estimation.

First, note that the Maximum Likelihood cost function (11) can be expressed as

$$\widetilde{Q}_{\mathrm{ML}}(\mathbf{p}_t) = \mathbf{v}_{\mathrm{ML}}^H(\mathbf{p}_t)\,\hat{\Gamma}\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t) \tag{16}$$

where

$$\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t) \triangleq \left[\left(\mathbf{w}_1^{\mathrm{ML}}\right)^T,\ldots,\left(\mathbf{w}_L^{\mathrm{ML}}\right)^T\right]^T \tag{17}$$

where the dependence of $\mathbf{w}_\ell^{\mathrm{ML}}$ on $\mathbf{p}_t$ is suppressed, $\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t)$ is a $(LM_rK) \times 1$ vector, and where $\hat{\Gamma}$ is a block diagonal matrix of dimensions $(LM_rK) \times (LM_rK)$ given by

$$\hat{\Gamma} \triangleq \begin{bmatrix}\hat{\mathbf{R}}_1 & \ldots & \mathbf{0}\\ \vdots & \ddots & \vdots\\ \mathbf{0} & \ldots & \hat{\mathbf{R}}_L\end{bmatrix} \tag{18}$$

Note that

$$\mathbf{v}_{\mathrm{ML}}^H(\mathbf{p}_t)\,\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t) = \sum_{\ell=1}^{L}\left(\mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t)\right)^H\mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t) = L \tag{19}$$

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where we used

$$\left(\mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t)\right)^H \mathbf{w}_\ell^{\mathrm{ML}}(\mathbf{p}_t) = (\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}})^H \mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}} = \bar{\mathbf{s}}^H \mathbf{A}_\ell^H(\mathbf{p}_t)\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}} = \bar{\mathbf{s}}^H\bar{\mathbf{s}} = 1. \tag{20}$$

The proposed MVDR weight vectors should satisfy

$$\mathbf{v}_{\mathrm{MVDR}} = \arg\min_{\mathbf{v}} \mathbf{v}^H \hat{\Gamma}\mathbf{v}, \tag{21}$$

subject to

$$\mathbf{v}^H \mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t) = L. \tag{22}$$

This is a minimization of a quadratic function under a linear constraint. This problem was solved in [16] using the complex gradient operator. The solution is given by

$$\mathbf{v}_{\mathrm{MVDR}}(\mathbf{p}_t) = L\frac{\hat{\Gamma}^{-1}\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t)}{\mathbf{v}_{\mathrm{ML}}^H(\mathbf{p}_t)\hat{\Gamma}^{-1}\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t)}. \tag{23}$$

The MVDR location estimator is therefore given by

$$\begin{aligned}\widehat{\mathbf{p}}_t^{\mathrm{MVDR}} &= \arg\max_{\mathbf{p}_t}\left[\mathbf{v}_{\mathrm{MVDR}}^H(\mathbf{p}_t)\hat{\Gamma}\mathbf{v}_{\mathrm{MVDR}}(\mathbf{p}_t)\right] \\ &= \arg\max_{\mathbf{p}_t}\frac{L^2\mathbf{v}_{\mathrm{ML}}^H(\mathbf{p}_t)\hat{\Gamma}^{-1}\hat{\Gamma}\hat{\Gamma}^{-1}\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t)}{\left[\mathbf{v}_{\mathrm{ML}}^H(\mathbf{p}_t)\hat{\Gamma}^{-1}\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t)\right]^2} \\ &= \arg\max_{\mathbf{p}_t}\frac{L^2}{\mathbf{v}_{\mathrm{ML}}^H(\mathbf{p}_t)\hat{\Gamma}^{-1}\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t)} \\ &= \arg\min_{\mathbf{p}_t}\tilde{Q}_{\mathrm{MVDR}}(\mathbf{p}_t)\end{aligned} \tag{24}$$

where we used $\left(\hat{\Gamma}^{-1}\right)^H = \hat{\Gamma}^{-1}$ and where

$$\tilde{Q}_{\mathrm{MVDR}}(\mathbf{p}_t) \triangleq \mathbf{v}_{\mathrm{ML}}^H(\mathbf{p}_t)\hat{\Gamma}^{-1}\mathbf{v}_{\mathrm{ML}}(\mathbf{p}_t). \tag{25}$$

This expression can be further simplified, using the fact that the inverse of $\hat{\Gamma}$ is a block diagonal matrix as well, given by

$$\hat{\Gamma}^{-1} \triangleq \begin{bmatrix}\hat{\mathbf{R}}_1^{-1} & \dots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \hat{\mathbf{R}}_L^{-1}\end{bmatrix} \tag{26}$$

Substituting (26) and (17) back in (25) we get

$$\tilde{Q}_{\mathrm{MVDR}}(\mathbf{p}_t) = \sum_{\ell=1}^{L} Q_\ell(\mathbf{p}_t) \tag{27}$$

where

$$Q_\ell(\mathbf{p}_t) \triangleq \bar{\mathbf{s}}^H \mathbf{A}_\ell^H(\mathbf{p}_t)\hat{\mathbf{R}}_\ell^{-1}\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}}. \tag{28}$$

This ends the derivation of the MVDR target location estimator. To summarize, the proposed estimator uses the raw data gathered at the sensors to evaluate the exact MVDR cost function of the target's position in a single step. Thus, MVDR is used for direct localization and not the estimation of the Direction of Arrival (DOA). It is important to note that no assumption was made on the number of targets during the derivation of the MVDR estimator. In comparison, an exact multi-target Maximum Likelihood estimator would need to know a-priori the number of targets, which is not always an easy task. Further, a mistake in the number of targets might cause severe degradation in its performance. Finally, the search dimension is N_t times that of the search dimension of the MVDR estimator developed here, where N_t is the number of targets. A possible algorithm is displayed in Algorithm 2.

Algorithm 2 Target Localization Using MVDR

Evaluate $\{\hat{\mathbf{R}}_\ell\}$ according to (12)
Define the area of interest and determine a suitable grid of locations $\mathbf{p}_1, \mathbf{p}_2 \dots \mathbf{p}_G$, where G is the number of grid points
for $j = 1$ to G **do**
  Set $Q(\mathbf{p}_j) = 0$
  for $\ell = 1$ to L **do**
    Evaluate $\tau_\ell(\mathbf{p}_j)$ according to (2)
    Evaluate $\mathbf{A}_\ell(\mathbf{p}_j)$ according to (4)
    Let $Q(\mathbf{p}_j) = Q(\mathbf{p}_j) + \bar{\mathbf{s}}^H \mathbf{A}_\ell^H \hat{\mathbf{R}}_\ell^{-1}\mathbf{A}_\ell\bar{\mathbf{s}}$
  end for
end for
Find the grid point for which Q is the smallest. This grid point is the estimated position.

C. Robustness of the Beamformer and Diagonal Loading

In practical scenarios the number of pulses M_p is usually small. In such cases the matrix $\hat{\mathbf{R}}_\ell$ is singular and cannot be inverted. We note that

$$\begin{aligned}\mathbf{R}_\ell &\triangleq E\left[\hat{\mathbf{R}}_\ell\right] \\ &= E\left[\alpha_\ell(j)\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t)\alpha_\ell^H(j)\right] + \sigma_n^2\mathbf{I} \\ &= \sigma_\alpha^2\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H(\mathbf{p}_t) + \sigma_n^2\mathbf{I}\end{aligned} \tag{29}$$

where σ_n^2 and σ_α^2 are, respectively, the noise and signal attenuation variances. Diagonal loading is a well known method (e.g. [17]–[19]) for inverting singular matrices. By replacing $\hat{\mathbf{R}}_\ell$ with

$$\hat{\mathbf{R}}_\ell = \hat{\mathbf{R}}_\ell^0 + \lambda\mathbf{I} \tag{30}$$

where $\hat{\mathbf{R}}_\ell^0$ is the original sample covariance matrix given by (12), the matrix can be inverted. We note that in radar systems, the noise variance σ_n^2 can be estimated by examining range cells distant from the cell-under-test, similar to the concept employed in constant-false-alarm-rate (CFAR) filters (see [20]). The effects of the level of diagonal loading λ were shown in [21] for the case of adaptive arrays. It was shown that diagonal loading reduces the effects of noise by increasing the noise-associated eigenvalues to the level of λ. While this has negligible effect on the nulling of interference represented by large eigenvalues, it may reduce the nulling capability of weak interference with small eigenvalues. Using the estimated noise variance, it is possible to choose the diagonal loading level according to the interference rejection needs. A good choice for λ is between the estimated noise level and the expected signal level.

III. Numerical Examples

In the following section we demonstrate the estimation and resolution capabilities of the proposed algorithm by

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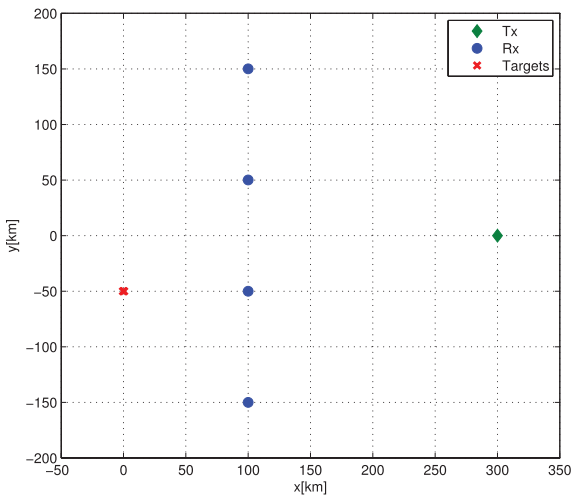

Fig. 1. Problem geometry for single element arrays layout. There are two targets present, but they are hard to discern due to the figure's scale.

numerical examples. We examine the algorithm for the case of multiple close targets. These targets may represent several separate targets, one target with several jamming signals, mimicking legitimate targets, or a single target with multiple resolvable scatterers, given that the system's parameters, such as the signal's bandwidth, provide such high resolution. We examine two possible layouts, the first consists of single element arrays ($M_r = 1$), and the second consists of arrays composed of several elements ($M_r > 1$). We emphasize that in both cases, the sensors themselves do not estimate TOA (Time of Arrival) or DOA (Direction of Arrival) parameters, but rather record the intercepted signals.

A. Single Element Arrays

The layout of the radar system used during simulations consists of a transmitter at $\{300, 0\}$[km] and four single element arrays distributed evenly from $\{100, 150\}$[km] to $\{100, -150\}$[km]. The problem geometry is shown in Fig. 1. This geometry is used in all the single-element arrays simulations.

The simulated signal sampling frequency is $f_s = 488$[kHz], the number of samples per pulse is $K = 128$ and the signal spectrum is flat with bandwidth of 150[kHz]. In order to gather enough statistics, the simulation results were based on 1000 Monte-Carlo runs. In each run the channel coefficients were drawn from a normal distribution given by $\boldsymbol{\alpha}_\ell \sim \mathcal{CN}(\mathbf{0}, \Lambda_\alpha)$ where $\boldsymbol{\alpha}_\ell \triangleq \left[\alpha_\ell(1), \ldots, \alpha_\ell(M_p)\right]^T$ and

$$\Lambda_\alpha \triangleq \begin{pmatrix} 1 & \rho & \ldots & \rho^{M_p-1} \\ \rho & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho \\ \rho^{M_p-1} & \ldots & \rho & 1 \end{pmatrix} \tag{31}$$

is a $M_p \times M_p$ matrix, and ρ is the correlation between the channel coefficients of adjacent pulses, equal to zero unless specifically stated otherwise. The noise samples were generated as complex-Gaussian random vectors as well, thus $\bar{\mathbf{n}}_\ell(j) \sim \mathcal{CN}\left(\mathbf{0}, \sigma_n^2 \mathbf{I}\right)$. The SNR is defined as

$$\begin{aligned} \mathrm{SNR} &\triangleq \frac{E\left\{\|\alpha_\ell(j)\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}}\|^2\right\}}{E\left\{\|\bar{\mathbf{n}}_\ell(j)\|^2\right\}} \\ &= \frac{E\left\{\|\alpha_\ell(j)\|^2\right\}\|\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}}\|^2}{K \cdot M_r \sigma_n^2} = \frac{1}{K \cdot M_r \sigma_n^2} \end{aligned} \tag{32}$$

where we used

$$\begin{aligned} E\left\{\|\alpha_\ell(j)\|^2\right\} &= 1 \\ \bar{\mathbf{s}}^H \mathbf{A}_\ell^H(\mathbf{p}_t)\mathbf{A}_\ell(\mathbf{p}_t)\bar{\mathbf{s}} &= \bar{\mathbf{s}}^H \bar{\mathbf{s}} = 1. \end{aligned} \tag{33}$$

Thus, the noise variance is given by

$$\sigma_n^2 = \frac{1}{K \cdot M_r \cdot \mathrm{SNR}}. \tag{34}$$

Unless stated otherwise, the number of pulses in all the simulations is $M_p = 2$. As discussed in Sec. II-C, in order to make the matrix $\hat{\mathbf{R}}_\ell$ invertible we use diagonal loading, where the diagonal loading level λ is chosen to be σ_n^2, motivated by the expected sample covariance matrix given in (29).

The simulations compare three positioning algorithms: the proposed MVDR method, the (single-target) Maximum Likelihood method, and a 2-step TOA-based localization method. We are interested in high resolution, so the TOA estimates for the 2-step method are obtained using MVDR, instead of traditional Maximum Likelihood based Matched Filters. This can be achieved by the same derivation in Sec. II-B and setting the number of receivers to one. The target's position is then given by minimizing the TOA cost function

$$Q_{\mathrm{TOA}}(\mathbf{p}_t) \triangleq \left[\hat{\mathbf{m}} - \mathbf{m}(\mathbf{p}_t)\right]^H \Sigma_{\mathrm{TOA}}^{-1}\left[\hat{\mathbf{m}} - \mathbf{m}(\mathbf{p}_t)\right] \tag{35}$$

where $\hat{\mathbf{m}}$ is a vector of L TOA measurements, L is the number of receivers, and where

$$\begin{aligned} \mathbf{m}(\mathbf{p}_t) &\triangleq \left[\tau_1(\mathbf{p}_t), \tau_2(\mathbf{p}_t), \ldots, \tau_L(\mathbf{p}_t)\right]^T \\ \Sigma_{\mathrm{TOA}} &\triangleq \mathrm{diag}\left\{\sigma_1^2, \sigma_2^2, \ldots, \sigma_L^2\right\} \end{aligned} \tag{36}$$

where $\tau_\ell(\mathbf{p}_t)$ is given by (2) and where σ_ℓ^2 denotes the variance of the ℓ-th TOA measurement, which is equal for all receivers in the simulated layout. Note that when there are multiple targets each receiver measures several TOAs. Finding the association of different TOAs at different receivers is generally not an easy task. In our simulations we avoid this issue by associating each measurement with the correct target. It should be emphasized that this is only an issue for 2-step algorithms.

Fig. 2 and Fig. 3 show the contour plots for the (single-target) Maximum Likelihood cost function defined in (11) and the inverse of the MVDR cost function defined in (27) for the case of two targets located at $\{-0.5, -50\}$[km] and $\{0.5, -50\}$[km] and no noise. It is clearly seen that the MVDR method succeeds in separating the targets whereas the ML method fails. In order to gain some insight into the difference between single-step and two-step methods, compare Fig. 2 and Fig. 3 to Fig. 4, which shows the contour plot for the inverse of the 2-step MVDR cost function defined in (35). Note that this shows the score of the second step only,

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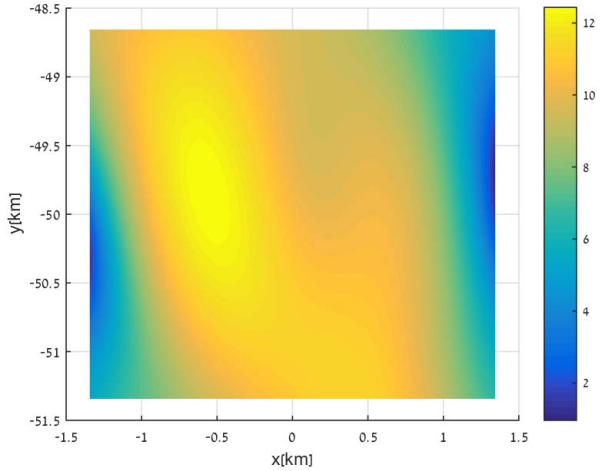

Fig. 2. Single element arrays layout: ML target position cost function in dB.

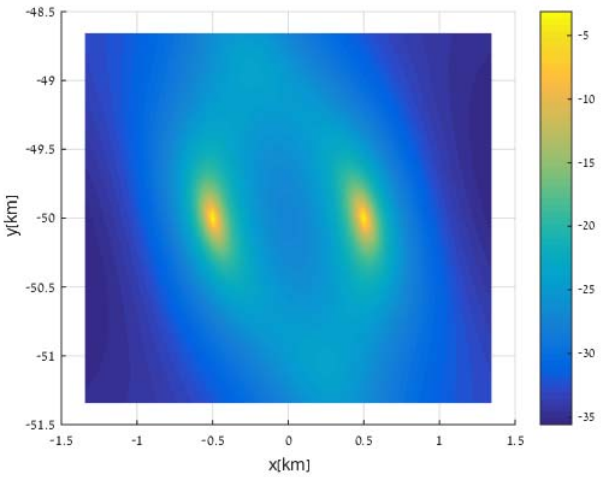

Fig. 3. Single element arrays layout: MVDR target position cost function in dB.

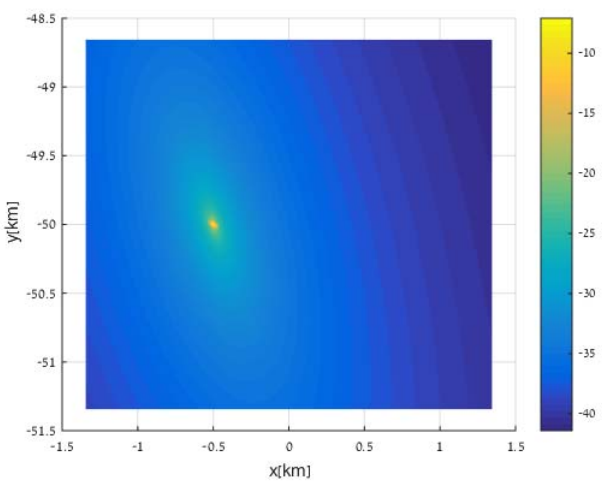

Fig. 4. Single element arrays layout: 2-step MVDR target position cost function in dB.

where the input to the second step is the TOA measurements which correspond to the target at $\{-0.5, -50\}$[km].

Next, we examine the effects of SNR on the estimation accuracy. Fig. 5 shows the location-root-mean-square-error (LRMSE) as a function of SNR for the three

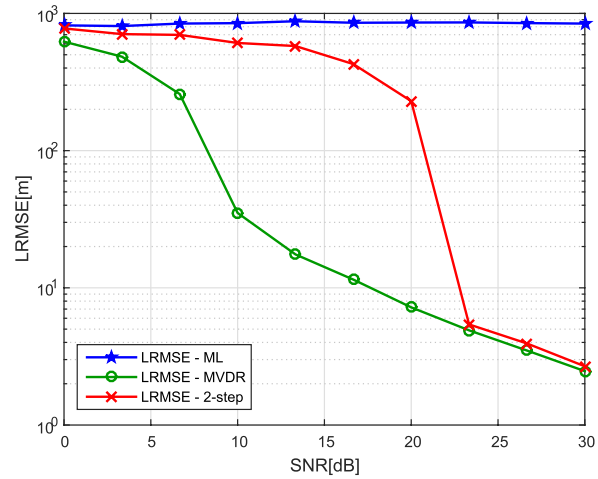

Fig. 5. Single element arrays layout: LRMSE as a function of SNR for two targets.

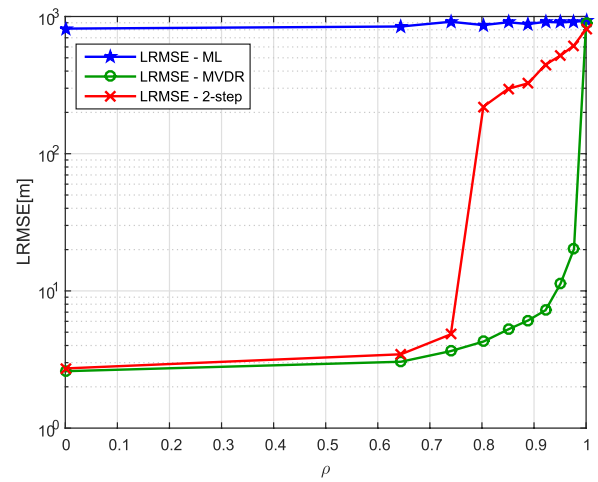

Fig. 6. Single element arrays layout: LRMSE as a function of ρ for two targets.

estimation techniques. The estimated targets' positions are the peaks closest to the actual targets' position of the corresponding cost functions. Thus, we ignore the problem of target detection, which could be addressed in future work. The LRMSE is evaluated separately for each target. However, performance for both targets are equivalent so only the LRMSE of one target is shown. Clearly, the single-target Maximum Likelihood does not improve with SNR while the MVDR shows excellent performance. The 2-step MVDR method converges to the 1-step approach, but is inferior at intermediate SNR.

It is interesting to examine the effect of the value of ρ presented in (31) on the LRMSE. In the previous section it was assumed that the signal attenuation is independent in each pulse, corresponding to $\rho = 0$. This assumption may not hold in some cases, e.g. for a slow moving, non-maneuvering target, or a target with relatively constant radar-cross-section (RCS). It is desirable to examine the impact of this dependency on the LRMSE. Fig. 6 shows the LRMSE for the previous case of two targets, as a function of the channel attenuation correlation ρ, for SNR $= 30$[dB].

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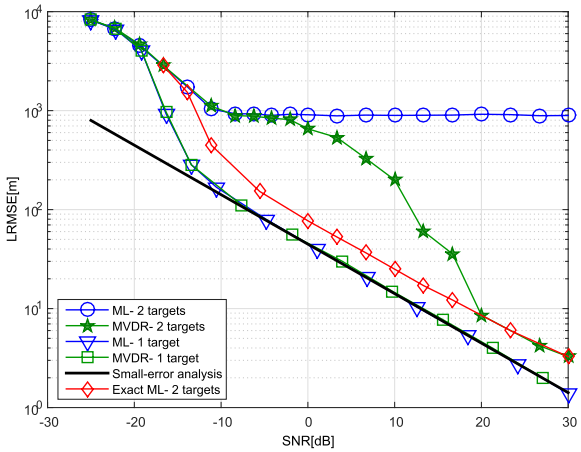

Fig. 7. Single element arrays layout: LRMSE as a function of SNR.

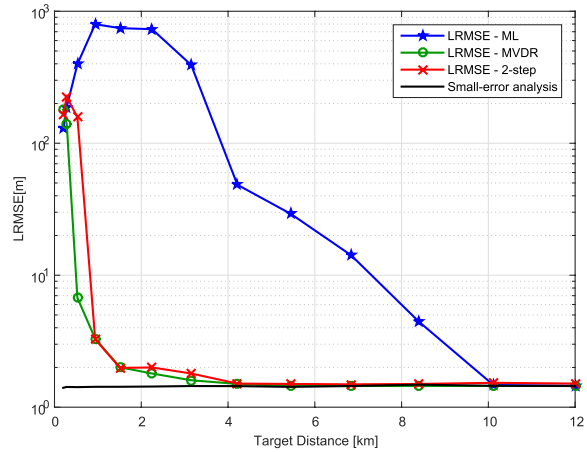

Fig. 8. Single element arrays layout: LRMSE as a function of the distance between targets.

It is shown that the LRMSE gradually increases with increasing ρ. However, this effect becomes critical only for a very high correlation of the channel attenuation, leading to the conclusion that the method proposed holds for practical radar scenarios. In comparison, the LRMSE for the 2-step method is less robust. We note that for $\rho = 1$, corresponding to a set of linearly dependent signals, the LRMSE using the proposed method is equivalent to the LRMSE using the Maximum Likelihood method. Note that the tolerability to signal attenuation correlation, coupled with the use of $M_p = 2$, enables the use of a small number of pulses with low pulse-repetition-interval (PRI) for processing, resulting in a short target dwell time, which in turn justifies the assumption that target speed can be considered negligible for the processing duration.

It is also of interest to examine the difference in performance of the estimation methods when only a single target is present. It is expected that in such a scenario the performance of the MVDR estimator would be inferior to the optimal single-target Maximum Likelihood estimator.

Fig. 7 shows the LRMSE as a function of SNR for the case of a single target located at $\{0, -50\}$[km]. The performance of the 2-step method are not shown in order to reduce clutter, since the superiority of the 1-step approach over the 2-step method for single target localization was already shown in [3]. The simulated performance and a theoretical small-error analysis for the MVDR estimator are all shown in the figure. The derivation of the small error performance expression is explained in the Appendix. Also shown in Fig. 7 is the performance for the previous case of two close targets. We note that both estimation techniques converge to the theoretical analysis and to one another at high SNR for a single target. This is a pleasing result, since the DPD method was shown in [3] to converge to the Cramer-Rao-Lower-Bound for this estimation problem. We also note that even for moderate SNR, both estimation methods yield similar results. Finally, note that the LRMSE for two close targets does not converge to the LRMSE for one target. This is because the two targets interfere with each other, causing an estimation bias. In order to provide a lower bound for the performance of the estimation, also shown in Fig. 7 is the LRMSE for the exact (multi-source) Maximum Likelihood. We emphasize again that the exact Maximum Likelihood has information which the other estimators do not, namely, it has to know a-priori the number of targets. In addition, it requires a multi-dimensional search, two major drawbacks when compared to the proposed method. In order to find the exact Maximum Likelihood we revisit (5). The received signal when several targets are present is given by

$$\bar{\mathbf{r}}_\ell(j) = \sum_{m=1}^{N_t} \alpha_{\ell,m}(j)\, \mathbf{A}_{\ell,m}(\mathbf{p}_m)\, \bar{\mathbf{s}} + \bar{\mathbf{n}}_\ell(j) \tag{37}$$

where N_t is the number of targets and $\mathbf{p}_m$ is the location of the m-th target. The exact Maximum Likelihood cost function is

$$Q(\{\mathbf{p}_m\}_{m=1,\ldots,N_t}) = \sum_{\ell=1}^{L}\sum_{j=1}^{M_p} \left\| \bar{\mathbf{r}}_\ell(j) - \sum_{m=1}^{N_t} \alpha_{\ell,m}(j)\, \mathbf{A}_{\ell,m}(\mathbf{p}_m)\, \bar{\mathbf{s}} \right\|^2 \tag{38}$$

Since $\{\alpha_{\ell,m}(j)\}$ are unknown deterministic constants, we can write

$$Q(\{\mathbf{p}_m\}_{m=1,\ldots,N_t}) = \sum_{\ell=1}^{L}\sum_{j=1}^{M_p} \min_{\{\alpha_{\ell,m}(j)\}_m} \left\| \bar{\mathbf{r}}_\ell(j) - \sum_{m=1}^{N_t} \alpha_{\ell,m}(j)\, \mathbf{A}_{\ell,m}(\mathbf{p}_m)\, \bar{\mathbf{s}} \right\|^2 \tag{39}$$

The exact Maximum Likelihood estimate of the set of targets' locations is given by minimizing (39) over all possible combinations of $\{\mathbf{p}_m\}$. The exact Maximum Likelihood performs better than the proposed method at intermediate SNR, which could be attributed to the extra information it possesses. However, at high SNR the MVDR converges to the exact ML which is the optimal estimator for this problem.

Next, it is interesting to examine what happens when the distance between the two targets is increased. It is expected that estimation performance will converge to the performance of a single target. Fig. 8 shows the LRMSE as a function of the targets separation d_p for SNR $= 30$[dB], where the targets

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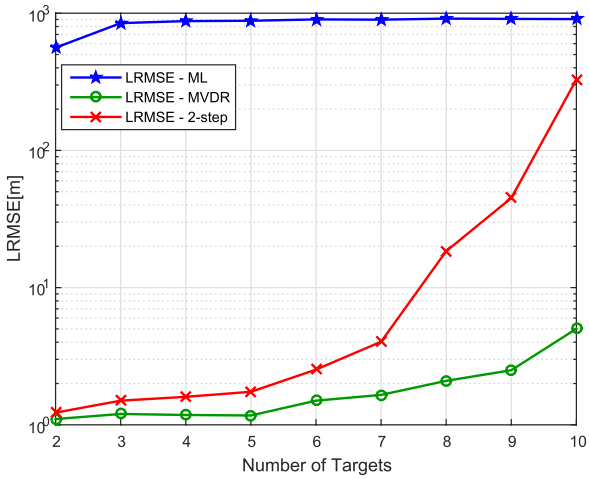

Fig. 9. Single element arrays layout: LRMSE as a function of the number of targets.

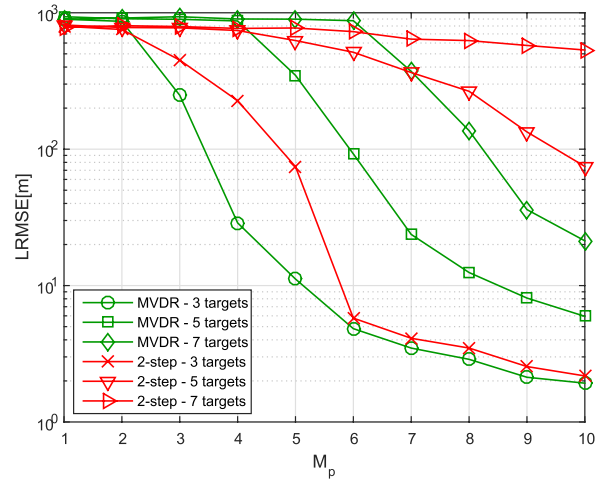

Fig. 10. Single element arrays layout: LRMSE as a function of the number of pulses.

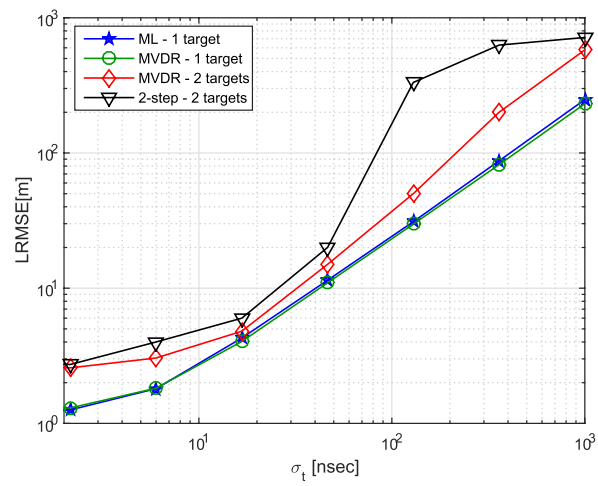

Fig. 11. Single element arrays layout: LRMSE as a function of the receivers' synchronization error standard deviation.

are at $\{-\frac{d_p}{2}, -50\}$[km] and $\{\frac{d_p}{2}, -50\}$[km]. Also shown is the theoretical small-error analysis for a single target at $\{-\frac{d_p}{2}, -50\}$[km]. For small target distance, all estimators fail to separate the targets, generally estimating a single position which is somewhere between the true targets' locations. Thus, as the distance is increased the LRMSE increases as well. This is true up to a point where the distance is high enough so that the targets are successfully separated. Further increase of the distance decreases the LRMSE, which finally converges to the small-error analysis for a single target. It is clearly seen that the MVDR method achieves target separation for closer targets than all other methods.

So far our layout consisted of at most two targets. Fig. 9 shows the LRMSE as a function of the number of targets present. The targets were placed along the x-axis around $\{0, -50\}$[km], spaced 1km from each other. The number of pulses in this simulation is $M_p = 32$, in order to suppress the effect of the linear dependence of the signals returned from different targets. It is evident that the single-step MVDR performs better than the other methods, especially for a large number of targets.

In order to complete our analysis of the performance for more than two targets, Fig. 10 shows the LRMSE as a function of the number of pulses for 3, 5 and 7 targets for the single-step and 2-step MVDR methods. As expected, when M_p is below the number of targets separation is not possible since the rank of the sample covariance matrix is lower than the number of targets. However, it is evident that the LRMSE improves rapidly when the number of pulses is increased further. It can be seen that the single-step MVDR performs better than the 2-step method for a smaller number of pulses.

So far we have assumed perfect synchronization between the receivers. We examine now the impact of receivers' synchronization errors on the estimators' performance. Fig. 11 shows the LRMSE of the estimators as a function of the synchronization error standard deviation σ_t for SNR $= 30$[dB] for a single target at $\{0, -50\}$[km] and for two targets at $\{-0.5, -50\}$[km] and $\{0.5, -50\}$[km]. In order to reduce clutter, the LRMSE of the 2-step method for a single target is not shown, since it is practically the same as the LRMSE of both other methods. Also not shown is the LRMSE of the ML method for two targets, since it was shown in Fig. 5 that it fails to separate the targets. It is seen that performance for all estimators for a single target are mostly the same, and that MVDR achieves target separation even for high synchronization errors. Note that synchronization errors smaller than $100n$sec can easily be achieved using, for example, GPS (e.g. [22]). Finally, we note that MVDR is more robust compared with the 2-step method.

B. Multi Element Receiving Arrays

We now turn to examine the case of receiving arrays ($M_r > 1$). Such arrays are usually harder to implement, as they require calibration of the array response. However, the additional data improves the LRMSE, possibly allowing the use of less arrays for target localization, depending on the problem geometry. To demonstrate this, the layout of this section consists of a transmitter at $\{300, 0\}$[km], and two receiving arrays, located at $\{50, 20\}$[km] and $\{50, -20\}$[km]. See Fig. 12. Five elements uniform linear arrays are used,

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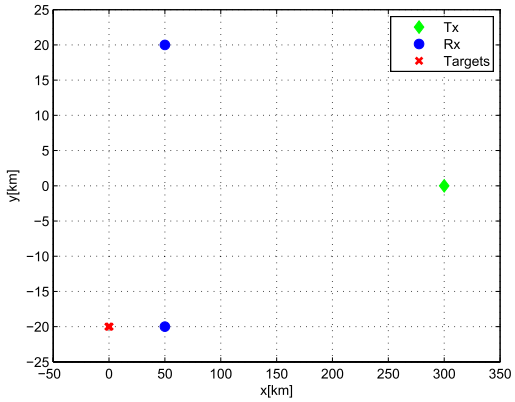

Fig. 12. Problem geometry for five elements arrays layout.

with half wavelength element spacing. The array baseline is parallel to the y axis. All other simulation parameters were unchanged.

Similarly to Sec. III-A, a 2-step based algorithm is used in the comparison. Here, in addition to measuring TOA, the sensors are also capable of DOA measurement, which was calculated using MVDR. We emphasize again that these measurements of intermediate parameters are only used for the 2-step approach. The target's position using this approach is given by minimizing the cost function

$$Q_{2-\text{step}}(\mathbf{p}_t) \triangleq Q_{\text{TOA}}(\mathbf{p}_t) + Q_{\text{DOA}}(\mathbf{p}_t) \tag{40}$$

where Q_{TOA} is given in (35) and

$$Q_{\text{DOA}}(\mathbf{p}_t) \triangleq \left[\hat{\mathbf{h}} - \mathbf{h}(\mathbf{p}_t)\right]^H \Sigma_{\text{DOA}}^{-1} \left[\hat{\mathbf{h}} - \mathbf{h}(\mathbf{p}_t)\right] \tag{41}$$

where $\hat{\mathbf{h}}$ is a vector of L DOA measurements, L is the number of receivers, and where

$$\begin{aligned} \mathbf{h}(\mathbf{p}_t) &\triangleq \left[\theta_1(\mathbf{p}_t), \theta_2(\mathbf{p}_t), \ldots, \theta_L(\mathbf{p}_t)\right]^T \\ \Sigma_{\text{DOA}} &\triangleq \text{diag}\left\{\varsigma_1^2, \varsigma_2^2, \ldots, \varsigma_L^2\right\} \end{aligned} \tag{42}$$

where θ_ℓ is the angle between the x axis and the line connecting the target at the hypothesized location $\mathbf{p}_t$ and the ℓ-th receiver and where ς_ℓ^2 denotes the variance of the ℓ-th DOA measurement. The variances of the DOA and TOA measurements were calculated numerically.

Fig. 13 and Fig. 14 show the contour plots for the Maximum Likelihood cost function defined in (11) and the inverse of the MVDR cost function defined in (27) for the case of two targets located at $\{-0.5, -20\}$[km] and $\{0.5, -20\}$[km] and no noise. The number of pulses is $M_p = 2$. Again, the MVDR method succeeds in separating the targets whereas the single target Maximum Likelihood method fails. For completeness, Fig. 15 shows the contour plot for the inverse of the 2-step MVDR cost function defined in (40). Remember that this is the second step of the localization, where the input for this function is the DOA and TOA measurements corresponding to the target at $\{-0.5, -20\}$[km].

Next, we show that the effects of SNR on the estimation accuracy are similar to the previous layout. Fig. 16 shows the LRMSE as a function of SNR for each target using MVDR, ML, and the 2-step MVDR method. We note that the addition of array elements allows for as few as two arrays to achieve high resolution target positioning. The single-step MVDR is again superior to the 2-step method at intermediate SNR.

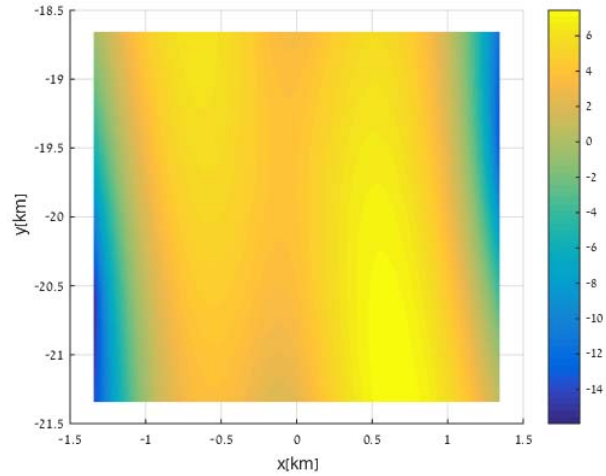

Fig. 13. Five elements arrays layout: ML target position cost function in dB.

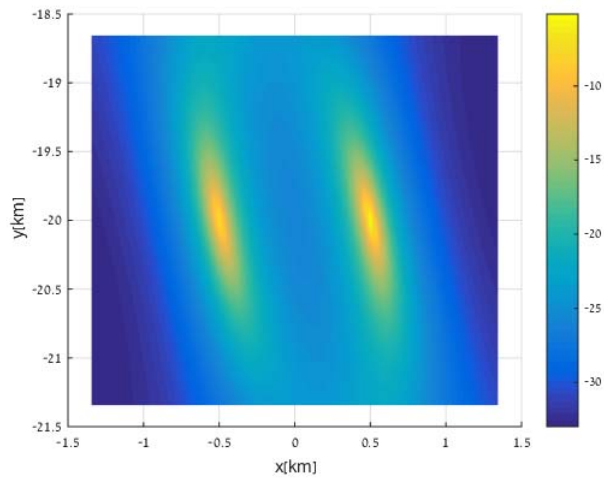

Fig. 14. Five elements arrays layout: MVDR target position cost function in dB.

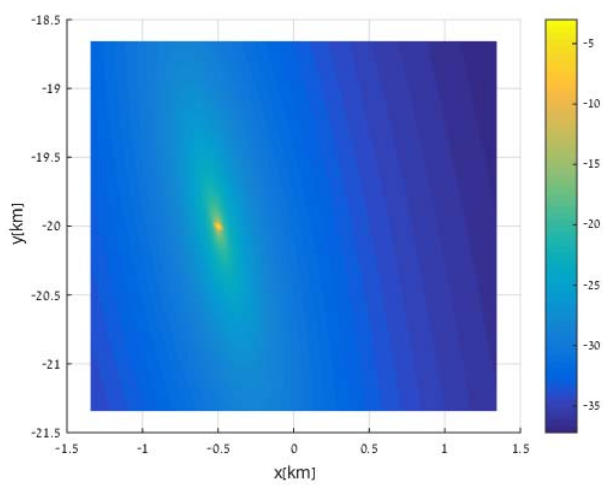

Fig. 15. Five elements arrays layout: 2-step MVDR target position cost function in dB.

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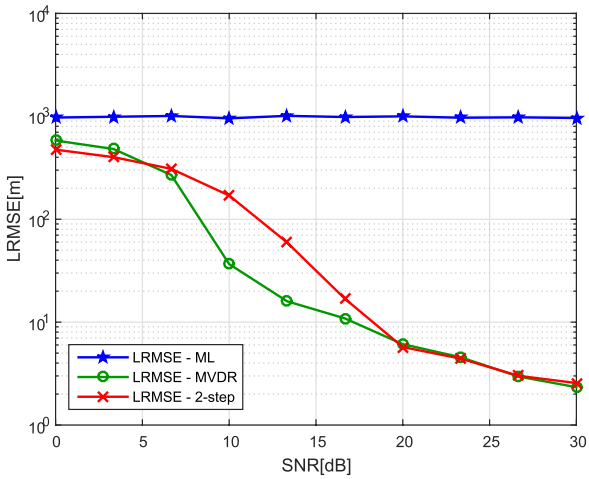

Fig. 16. Five elements arrays layout: LRMSE as a function of SNR for two targets.

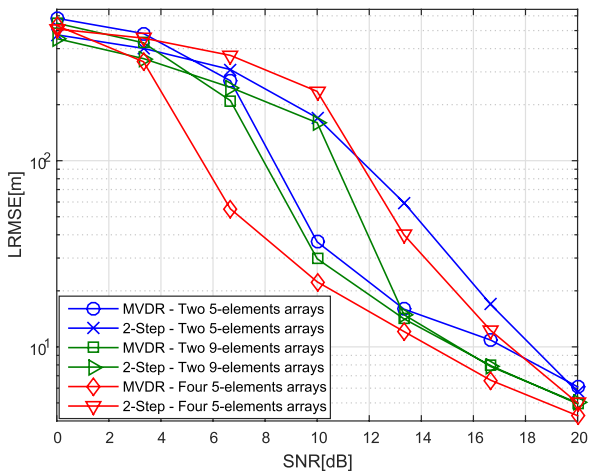

Fig. 17. Multi elements arrays layout: LRMSE as a function of SNR for two targets.

In order to gain some intuitive understanding of the difference between the performance of the single-step and 2-step MVDR for multiple targets, Fig. 17 shows the LRMSE for the three following layouts. 1) The previous 5-element receiving arrays layout. 2) Two 9-element receiving arrays at $\{50, 20\}$[km] and $\{50, -20\}$[km]. 3) Four 5-element receiving arrays at $\{50, 40\}$[km], $\{50, 20\}$[km], $\{50, -20\}$[km] and $\{50, -40\}$[km]. All other parameters are unchanged. It is evident that the single-step MVDR is consistently superior to the 2-step method.

IV. Conclusion

In this paper we proposed a single-step direct position determination (DPD) using the MVDR approach rather than the single target Maximum Likelihood approach. The proposed method is an adaptive method, using the returns from several consecutive pulses for computation of the target's location directly without first estimation of direction of arrival and delay. We presented DPD based on single target Maximum Likelihood and DPD based on MVDR for multistatic radar, but our approach may easily be extended to other localization problems. We demonstrated that the MVDR approach achieves superior resolution with respect to the Maximum Likelihood and to current 2-step MVDR based localization methods by analyzing multiple near targets localization. The targets used in the analysis can represent a single target with multiple, resolvable scatterers or represent separate, small targets. The fine resolution can be employed to resolving real targets from many decoys. We have demonstrated localization using a small number of pulses, and have shown that returns of consecutive pulses can be highly correlative, allowing the use of this method in most practical scenarios. Finally, we've developed a way to predict estimation performance for a single target at high SNR based on small-error analysis.

Appendix
Small Error Analysis

We derive the covariance matrix of the MVDR position error, assuming a single target and small errors. The covariance can be used to predict estimation performance for high SNR or many pulses. The following could easily be derived for the ML estimator as well. However, the Cramer-Rao-Lower-Bound for this estimation problem was derived in [3] and it was shown that the DPD method converges to it. The framework and formulas used for the complex calculus performed here were presented in [23]. For the purpose of a compact notation, throughout the appendix we use $\mathbf{w}_\ell$ instead of $\mathbf{w}_\ell^{\mathrm{ML}}$ defined in (15), and the dependence on $\mathbf{p}_t$ will be suppressed.

We begin our derivation by defining

$$\begin{aligned}\Delta\mathbf{p}_t &\triangleq \hat{\mathbf{p}}_t - \mathbf{p}_t\\ \bar{\mathbf{r}}_{0,\ell}(j) &\triangleq \alpha_\ell(j)\mathbf{A}_\ell\bar{\mathbf{s}}\\ \Delta\bar{\mathbf{r}}_\ell(j) &\triangleq \bar{\mathbf{r}}_\ell(j) - \bar{\mathbf{r}}_{0,\ell}(j) = \bar{\mathbf{n}}_\ell(j)\end{aligned} \tag{43}$$

Using the Woodbury identity we get

$$\begin{aligned}\hat{\mathbf{R}}_\ell^{-1}\Big|_{\{\bar{\mathbf{r}}_{0,\ell}(j)\}} &= \left(\lambda\mathbf{I} + \widehat{\alpha}_\ell\mathbf{A}_\ell\bar{\mathbf{s}}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H\right)^{-1}\\ &= \frac{1}{\lambda}\left(\mathbf{I} - \frac{\mathbf{A}_\ell\bar{\mathbf{s}}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H}{\bar{\mathbf{s}}^H\mathbf{A}_\ell^H\mathbf{A}_\ell\bar{\mathbf{s}} + \lambda/\widehat{\alpha}_\ell}\right)\\ &= \frac{1}{\lambda}\left(\mathbf{I} - \frac{\mathbf{A}_\ell\bar{\mathbf{s}}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H}{1 + \lambda/\widehat{\alpha}_\ell}\right)\end{aligned} \tag{44}$$

where λ is the sample covariance matrix loading factor defined in (30), and

$$\widehat{\alpha}_\ell \triangleq \frac{1}{M_p}\sum_{j=1}^{M_p}|\alpha_\ell(j)|^2 \tag{45}$$

We note that

$$\begin{aligned}\widetilde{Q}_{\mathrm{MVDR}}(\mathbf{p}_t)\Big|_{\{\bar{\mathbf{r}}_{0,\ell}(j)\}_{\ell,j}} &= \sum_{\ell=1}^{L}\bar{\mathbf{s}}^H\hat{\mathbf{A}}_\ell^H\hat{\mathbf{R}}_\ell^{-1}\Big|_{\{\bar{\mathbf{r}}_{0,\ell}(j)\}}\hat{\mathbf{A}}_\ell\bar{\mathbf{s}}\\ &= \sum_{\ell=1}^{L}\bar{\mathbf{s}}^H\hat{\mathbf{A}}_\ell^H\frac{1}{\lambda}\left(\mathbf{I} - \frac{\mathbf{A}_\ell\bar{\mathbf{s}}\bar{\mathbf{s}}^H\mathbf{A}_\ell^H}{1 + \lambda/\widehat{\alpha}_\ell}\right)\hat{\mathbf{A}}_\ell\bar{\mathbf{s}}\\ &= \frac{1}{\lambda}\sum_{\ell=1}^{L}\left(1 - \frac{\left|\bar{\mathbf{s}}^H\hat{\mathbf{A}}_\ell^H\mathbf{A}_\ell\bar{\mathbf{s}}\right|^2}{1 + \lambda/\widehat{\alpha}_\ell}\right)\end{aligned} \tag{46}$$

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where we used (44) for the second transition and (8) for the third transition. It is obvious that minimization of (46) is achieved by choosing $\hat{\mathbf{p}}_t = \mathbf{p}_t$. This can be understood in the following manner: the attenuation coefficient does not affect the estimated position when there is no noise present. Since this choice also maximizes the individual cost functions comprising $\widetilde{Q}_{\text{MVDR}}$, we get

$$\left.\frac{\partial Q_\ell}{\partial \mathbf{p}_t}\right|_{\hat{\mathbf{p}}_t,\{\bar{\mathbf{r}}(j)\}} = \left.\frac{\partial Q_\ell}{\partial \mathbf{p}_t}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} = 0 \tag{47}$$

where we used the more compact notation

$$(\cdot)|_{\{\bar{\mathbf{r}}_0(j)\}} \triangleq (\cdot)|_{\{\bar{\mathbf{r}}_{0,\ell}(j)\}_{\ell,j}}$$

The first order Taylor series of $\frac{\partial Q_\ell}{\partial \mathbf{p}_t}$ is given by

$$\begin{aligned}\left.\frac{\partial Q_\ell}{\partial \mathbf{p}_t}\right|_{\hat{\mathbf{p}}_t,\{\bar{\mathbf{r}}(j)\}} &= \left.\frac{\partial Q_\ell}{\partial \mathbf{p}_t}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} + \left.\frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t^2}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} \Delta\mathbf{p}_t \\ &\quad + 2\sum_{j=1}^{M_p} \text{Re}\left\{ \left.\frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t \partial \bar{\mathbf{r}}_\ell(j)}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} \Delta\bar{\mathbf{r}}_\ell(j)\right\}\end{aligned} \tag{48}$$

where we used the fact that $\frac{\partial Q_\ell}{\partial \mathbf{p}_t}$ is a real function. Using

$$\left.\frac{\partial \tilde{Q}_{\text{MVDR}}}{\partial \mathbf{p}_t}\right|_{\hat{\mathbf{p}}_t,\{\bar{\mathbf{r}}(j)\}} = \sum_{\ell=1}^{L} \left.\frac{\partial Q_\ell}{\partial \mathbf{p}_t}\right|_{\hat{\mathbf{p}}_t,\{\bar{\mathbf{r}}(j)\}} \tag{49}$$

and (47) we get

$$\begin{aligned}\Delta\mathbf{p}_t = -2&\left(\sum_{\ell=1}^{L} \left.\frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t^2}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}}\right)^{-1} \\ &\cdot \sum_{\ell=1}^{L}\sum_{j=1}^{M_p} \text{Re}\left\{ \left.\frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t \partial \bar{\mathbf{r}}_\ell(j)}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} \Delta\bar{\mathbf{r}}_\ell(j)\right\}.\end{aligned} \tag{50}$$

Note that

$$\begin{aligned}\text{E}\left[\Delta\bar{\mathbf{r}}_{\ell_1}(j_1)\,\Delta\bar{\mathbf{r}}_{\ell_2}^T(j_2)\right] &= 0 \\ \text{E}\left[\Delta\bar{\mathbf{r}}_{\ell_1}(j_1)\,\Delta\bar{\mathbf{r}}_{\ell_2}^H(j_2)\right] &= 0 \\ \text{E}\left[\Delta\bar{\mathbf{r}}_{\ell}(j)\,\Delta\bar{\mathbf{r}}_{\ell}^T(j)\right] &= 0 \\ \text{E}\left[\Delta\bar{\mathbf{r}}_{\ell}(j)\,\Delta\bar{\mathbf{r}}_{\ell}^H(j)\right] &\triangleq \Lambda_n = \sigma_n^2 \mathbf{I}_{K\cdot M_r}\end{aligned} \tag{51}$$

and thus we get

$$\begin{aligned}\mathbf{C}_{\mathbf{p},\mathbf{p}} &\triangleq \text{E}\left[\Delta\mathbf{p}_t \Delta\mathbf{p}_t^T\right] \\ &\cong 2\sigma_n^2\left(\sum_{\ell=1}^{L} \left.\frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t^2}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}}\right)^{-1} \\ &\quad \cdot \text{Re}\left\{\sum_{\ell=1}^{L}\sum_{j=1}^{M_p} \frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t \partial \bar{\mathbf{r}}_\ell(j)}\left(\frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t \partial \bar{\mathbf{r}}_\ell(j)}\right)^H\right\}\Bigg|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} \\ &\quad \cdot \left(\sum_{\ell=1}^{L} \left.\frac{\partial^2 Q_\ell}{\partial \mathbf{p}_t^2}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}}\right)^{-T}.\end{aligned} \tag{52}$$

We now turn to find each of the partial derivatives constituting the above expression.

In the next section, the subscript $(\cdot)_\ell$ is omitted whenever the discussion is focused on a single receiver-transmitter pair.

We begin with an expression for $\mathbf{w}_{\mathbf{p}}^T \triangleq \frac{\partial \mathbf{w}}{\partial \mathbf{p}_t}$. In order to achieve compact expressions we assume each receiver consists of a single element ($M_r = 1$). It is possible, however, to develop expressions for the case of a general array response. Define the quantities

$$\begin{aligned}\mathbf{p}_t &\triangleq (x, y)^T \\ \mathbf{p}_\ell &\triangleq (x_\ell, y_\ell)^T \\ \mathbf{p}_{\text{Tx}} &\triangleq (x_{\text{Tx}}, y_{\text{Tx}})^T \\ \mathbf{d}_\ell &\triangleq [x - x_\ell, y - y_\ell]^T \\ \mathbf{d}_{\text{Tx}} &\triangleq [x - x_{\text{Tx}}, y - y_{\text{Tx}}]^T\end{aligned} \tag{53}$$

where $\mathbf{p}_{\text{Tx}}$ and $\mathbf{p}_\ell$ are, respectively, the transmitter's and ℓ-th receiver's locations. The signal delay is given by

$$\begin{aligned}\tau_\ell &= \frac{\|\mathbf{p}_t - \mathbf{p}_\ell\|}{c} + \frac{\|\mathbf{p}_t - \mathbf{p}_{\text{Tx}}\|}{c} \\ &= \frac{\sqrt{(x - x_\ell)^2 + (y - y_\ell)^2}}{c} + \frac{\sqrt{(x - x_{\text{Tx}})^2 + (y - y_{\text{Tx}})^2}}{c}.\end{aligned} \tag{54}$$

The partial derivative with respect to the target coordinate x is given by

$$\begin{aligned}\frac{\partial \tau_\ell}{\partial x} &= \frac{x - x_\ell}{c\sqrt{(x - x_\ell)^2 + (y - y_\ell)^2}} \\ &\quad + \frac{x - x_{\text{Tx}}}{c\sqrt{(x - x_{\text{Tx}})^2 + (y - y_{\text{Tx}})^2}}\end{aligned} \tag{55}$$

Define the $K \times K$ matrix

$$\mathbf{F} \triangleq \text{diag}\,(-j2\pi f_1, \ldots, -j2\pi f_K) \tag{56}$$

Using the definition in (15) we get

$$\frac{\partial \mathbf{w}_\ell}{\partial x} = \frac{\partial \mathbf{A}_\ell}{\partial x}\bar{\mathbf{s}} = \left[\frac{x - x_\ell}{c\,\|\mathbf{d}_\ell\|} + \frac{x - x_{\text{Tx}}}{c\,\|\mathbf{d}_{\text{Tx}}\|}\right]\mathbf{F}\mathbf{A}_\ell\bar{\mathbf{s}} \tag{57}$$

which can be expressed in vector form, combining it with the partial derivative with respect to the target coordinate y, by the $2 \times K$ matrix $\mathbf{w}_{\mathbf{p}}$, given by

$$\mathbf{w}_{\mathbf{p}}^T = \frac{1}{c}\mathbf{F}\mathbf{w}_\ell \check{\mathbf{d}}_\ell^T \tag{58}$$

where

$$\check{\mathbf{d}}_\ell \triangleq \frac{\mathbf{d}_\ell}{\|\mathbf{d}_\ell\|} + \frac{\mathbf{d}_{\text{Tx}}}{\|\mathbf{d}_{\text{Tx}}\|} = [\cos\theta_\ell + \cos\phi_{\text{Tx}}, \sin\theta_\ell + \sin\phi_{\text{Tx}}]^T \tag{59}$$

θ_ℓ is the angle between the x axis and the line connecting the target and the ℓ-th receiver. ϕ_{Tx} is the angle between the x axis and the line connecting the target and the transmitter.

Next, we derive the expressions for $\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{p}_t}$ and $\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \bar{\mathbf{r}}(j)}$, where $\mathbf{q}_{\mathbf{p}}^T \triangleq \frac{\partial Q}{\partial \mathbf{p}_t}$. Note that the individual cost functions defined in (28) can be expressed as

$$Q_\ell = \mathbf{w}_\ell^H \hat{\mathbf{R}}_\ell^{-1} \mathbf{w}_\ell \tag{60}$$

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Using the chain rule, we get

$$\begin{aligned}\mathbf{q}_{\mathbf{p}}^T &\triangleq \frac{\partial Q}{\partial \mathbf{p}_t} = \frac{\partial Q}{\partial \mathbf{w}}\frac{\partial \mathbf{w}}{\partial \mathbf{p}_t} + \frac{\partial Q}{\partial \mathbf{w}^*}\frac{\partial \mathbf{w}^*}{\partial \mathbf{p}_t} \\ &= \mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{w}_{\mathbf{p}}^T + \mathbf{w}^T\hat{\mathbf{R}}^{-T}\mathbf{w}_{\mathbf{p}}^H \\ &= \frac{2}{c}\mathrm{Re}\left\{\mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w}\right\}\check{\mathbf{d}}_\ell^T \end{aligned} \tag{61}$$

where we used (58), and where $\frac{\partial f}{\partial z}$ is defined as the partial derivative of f with respect to z when z^* is treated as a constant, where z^* is the complex conjugate of z. Similarly, $\frac{\partial f}{\partial z^*}$ is defined as the partial derivative of f with respect to z^* when z is treated as a constant. (61) can also be expressed as

$$\mathbf{q}_{\mathbf{p}} = \frac{1}{c}\left(\mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w} + \mathbf{w}^T\hat{\mathbf{R}}^{-T}\mathbf{F}^*\mathbf{w}^*\right)\check{\mathbf{d}}_\ell \tag{62}$$

The partial derivatives of $\mathbf{q}_{\mathbf{p}}$ with respect to $\mathbf{w}$ and $\mathbf{w}^*$ are given by the $2 \times K$ matrices

$$\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{w}} = \frac{1}{c}\check{\mathbf{d}}_\ell\left(\mathbf{w}^H\mathbf{F}^H\hat{\mathbf{R}}^{-1} + \mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{F}\right) \tag{63}$$

and

$$\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{w}^*} = \frac{1}{c}\check{\mathbf{d}}_\ell\left(\mathbf{w}^T\mathbf{F}\hat{\mathbf{R}}^{-T} + \mathbf{w}^T\hat{\mathbf{R}}^{-T}\mathbf{F}^*\right) \tag{64}$$

Using the chain rule and some simple algebraic manipulations, we get

$$\begin{aligned}\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{p}_t} &= \frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{w}}\mathbf{w}_{\mathbf{p}}^T + \frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{w}^*}\mathbf{w}_{\mathbf{p}}^H \\ &= \frac{2}{c^2}\check{\mathbf{d}}_\ell\mathrm{Re}\left\{\mathbf{w}^H\mathbf{F}^*\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w} + \mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{F}\mathbf{w}\right\}\check{\mathbf{d}}_\ell^T \end{aligned} \tag{65}$$

where $\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{p}_t}$ is a matrix of dimensions 2×2.

Next, we derive the partial derivative of $\mathbf{q}_{\mathbf{p}}$ with respect to the received signal $\bar{\mathbf{r}}(j)$. The partial derivative of (62) with respect to the g-th element of the received signal, denoted by $\bar{\mathbf{r}}_g(j)$, is given by

$$\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \bar{\mathbf{r}}_g(j)} = \frac{1}{c}\left[\frac{\partial\left(\mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w}\right)}{\partial \bar{\mathbf{r}}_g(j)} + \frac{\partial\left(\mathbf{w}^T\hat{\mathbf{R}}^{-T}\mathbf{F}^*\mathbf{w}^*\right)}{\partial \bar{\mathbf{r}}_g(j)}\right]\check{\mathbf{d}}_\ell \tag{66}$$

The partial derivatives in (66) are given by

$$\begin{aligned}\frac{\partial\left(\mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w}\right)}{\partial \bar{\mathbf{r}}_g(j)} &= -\mathbf{w}^H\hat{\mathbf{R}}^{-1}\frac{\partial \hat{\mathbf{R}}}{\partial \bar{\mathbf{r}}_g(j)}\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w} \\ &= -\frac{1}{M_p}\mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{e}_g\bar{\mathbf{r}}^H(j)\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w} \end{aligned} \tag{67}$$

and

$$\begin{aligned}\frac{\partial\left(\mathbf{w}^T\hat{\mathbf{R}}^{-T}\mathbf{F}^*\mathbf{w}^*\right)}{\partial \bar{\mathbf{r}}_g(j)} &= \mathrm{conj}\left\{\frac{\partial\left(\mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w}\right)}{\partial \bar{\mathbf{r}}_g(j)}\right\} \\ &= -\frac{1}{M_p}\mathbf{e}_g^T\hat{\mathbf{R}}^{-T}\mathbf{F}^*\mathbf{w}^*\mathbf{w}^T\hat{\mathbf{R}}^{-T}\bar{\mathbf{r}}^*(j) \end{aligned} \tag{68}$$

where $\mathbf{e}_g$ is a column vector where each element is 0 except for the g-th element, equal to 1. Substituting (67) and (68) back in (66) we get

$$\begin{aligned}&\left.\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \bar{\mathbf{r}}_g(j)}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} \\ &= -\frac{1}{c\cdot M_p}\check{\mathbf{d}}_\ell\begin{bmatrix}\bar{\mathbf{r}}^H(j)\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w}\mathbf{w}^H\hat{\mathbf{R}}^{-1}\mathbf{e}_g \\ +\mathbf{e}_g^T\hat{\mathbf{R}}^{-T}\mathbf{F}^*\mathbf{w}^*\mathbf{w}^T\hat{\mathbf{R}}^{-T}\bar{\mathbf{r}}^*(j)\end{bmatrix} \\ &= -\frac{1}{c\cdot M_p}\check{\mathbf{d}}_\ell\begin{bmatrix}\bar{\mathbf{r}}^H(j)\hat{\mathbf{R}}^{-1}\mathbf{F}\mathbf{w}\mathbf{w}^H\hat{\mathbf{R}}^{-1} \\ +\bar{\mathbf{r}}^H(j)\hat{\mathbf{R}}^{-1}\mathbf{w}\mathbf{w}^H\mathbf{F}^*\hat{\mathbf{R}}^{-1}\end{bmatrix}\mathbf{e}_g \end{aligned} \tag{69}$$

where we used the fact that the terms in the brackets are scalars. We can now write (69) in vector form

$$\begin{aligned}&\left.\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \bar{\mathbf{r}}(j)}\right|_{\mathbf{p}_t,\{\bar{\mathbf{r}}_0(j)\}} \\ &= -\frac{1}{c\cdot M_p}\check{\mathbf{d}}_\ell\bar{\mathbf{r}}^H(j)\hat{\mathbf{R}}^{-1}\left[\mathbf{F}\mathbf{w}\mathbf{w}^H + \mathbf{w}\mathbf{w}^H\mathbf{F}^*\right]\hat{\mathbf{R}}^{-1} \end{aligned} \tag{70}$$

where $\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \bar{\mathbf{r}}(j)}$ is a $2 \times K$ matrix. To summarize, using (52), (65) and (70) one can calculate $\mathbf{C}_{\mathbf{p},\mathbf{p}}$ and the LRMSE for the proposed method for high SNR.

In order to obtain some insight we examine this analysis when the signal attenuation is constant for all pulses. This condition can be expressed, without loss of generality, as $\alpha_\ell(j) = 1$ for all ℓ and j. Note that under this condition we get

$$\begin{aligned}\bar{\mathbf{r}}(j) &= \mathbf{w} \\ \hat{\mathbf{R}} &= \mathbf{w}\mathbf{w}^H \end{aligned} \tag{71}$$

Substituting (71) and (44) in (65) we get

$$\begin{aligned}\frac{\partial \mathbf{q}_{\mathbf{p}}}{\partial \mathbf{p}_t} &= \frac{2}{\lambda c^2}\check{\mathbf{d}}_\ell\cdot\mathrm{Re}\left\{\begin{matrix}\mathbf{w}^H\mathbf{F}^*\mathbf{F}\mathbf{w} - \frac{1}{1+\lambda}\mathbf{w}^H\mathbf{F}^*\mathbf{w}\mathbf{w}^H\mathbf{F}\mathbf{w} \\ +\frac{\lambda}{1+\lambda}\mathbf{w}^H\mathbf{F}\mathbf{F}\mathbf{w}\end{matrix}\right\}\check{\mathbf{d}}_\ell^T \\ &\cong -\frac{2}{\lambda c^2}\check{\mathbf{d}}_\ell\left(\eta^2 - \beta^2\right)\check{\mathbf{d}}_\ell^T \end{aligned} \tag{72}$$

where we used $\lambda \to 0$. and where

$$\begin{aligned}\eta &\triangleq \sum_{k=1}^K 2\pi f_k|w_\ell(k)|^2 = \sum_{k=1}^K 2\pi f_k|\bar{s}_k|^2 \\ \beta &\triangleq \sqrt{\sum_{k=1}^K (2\pi f_k)^2|w_\ell(k)|^2} = \sqrt{\sum_{k=1}^K (2\pi f_k)^2|\bar{s}_k|^2} \end{aligned} \tag{73}$$

and $w_\ell(k)$ is the k-th element of $\mathbf{w}_\ell$, and where we used

$$|w_\ell(k)|^2 = |\bar{s}_k e^{-j2\pi f_k\tau_\ell}|^2 = |\bar{s}_k|^2 \tag{74}$$

Define

$$\gamma \triangleq \sqrt{\sum_{k=1}^K (2\pi f_k - \eta)^2|\bar{s}_k|^2} \tag{75}$$

which can be expressed as

$$\gamma^2 = \beta^2 - 2\eta\sum_{k=1}^K 2\pi f_k|\bar{s}_k|^2 + \eta^2\bar{\mathbf{s}}^H\bar{\mathbf{s}} = \beta^2 - \eta^2 \tag{76}$$

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Note that η and γ can be understood as the central frequency and effective bandwidth of $\bar{\mathbf{s}}$, respectively. Next, substituting (71) and (44) in (70) we get

$$\begin{aligned}\frac{\partial \mathbf{q_p}}{\partial \bar{\mathbf{r}}(j)} &= -\frac{1}{cM_p\lambda\,(1+\lambda)}\check{\mathbf{d}}_\ell \mathbf{w}^H \\ &\quad \cdot \begin{bmatrix} \mathbf{F}\mathbf{w}\mathbf{w}^H + \mathbf{w}\mathbf{w}^H\mathbf{F}^* \\ -\frac{1}{1+\lambda}\left(\mathbf{F}\mathbf{w}\mathbf{w}^H + \mathbf{w}\mathbf{w}^H\mathbf{F}^*\mathbf{w}\mathbf{w}^H\right) \end{bmatrix} \\ &\cong -\frac{1}{cM_p\lambda}\check{\mathbf{d}}_\ell\left(\mathbf{w}^H\mathbf{F}^* - \mathbf{w}^H\mathbf{F}^*\mathbf{w}\mathbf{w}^H\right) \end{aligned} \tag{77}$$

Note that

$$\begin{aligned}&\frac{\partial \mathbf{q_p}}{\partial \bar{\mathbf{r}}(j)}\left[\frac{\partial \mathbf{q_p}}{\partial \bar{\mathbf{r}}(j)}\right]^H \\ &\cong \frac{1}{\left(cM_p\lambda\right)^2}\check{\mathbf{d}}_\ell\left(\mathbf{w}^H\mathbf{F}^*\mathbf{F}\mathbf{w} - \mathbf{w}^H\mathbf{F}^*\mathbf{w}\mathbf{w}^H\mathbf{F}\mathbf{w}\right)\check{\mathbf{d}}_\ell^T \\ &= \frac{1}{\left(cM_p\lambda\right)^2}\check{\mathbf{d}}_\ell\left(\beta^2 - \eta^2\right)\check{\mathbf{d}}_\ell^T \end{aligned} \tag{78}$$

Finally, substituting (72) and (78) in (52) we get

$$\mathbf{C_{p,p}} \cong \frac{\sigma_n^2\ c^2}{2\ M_p\gamma^2}\left[\sum_{\ell=1}^{L}\check{\mathbf{d}}_\ell\check{\mathbf{d}}_\ell^T\right]^{-1} \tag{79}$$

Furthermore, using (59) we get

$$\check{\mathbf{d}}_\ell\check{\mathbf{d}}_\ell^T = \begin{bmatrix} u_\ell^2 & u_\ell v_\ell \\ u_\ell v_\ell & v_\ell^2 \end{bmatrix} \tag{80}$$

where

$$\begin{aligned} u_\ell &\triangleq \cos\theta_\ell + \cos\phi_{\mathrm{Tx}} \\ v_\ell &\triangleq \sin\theta_\ell + \sin\phi_{\mathrm{Tx}} \end{aligned} \tag{81}$$

Delving into (79), we note that the estimation error is inversely proportional to the number of pulses and to the effective bandwidth of the transmitted signal. Also note that the error is independent of transmitter-target range and the target-receivers ranges.

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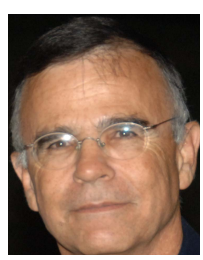

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